

Perfume Notes and Synesthetic Colors

A data-science study: do colors that match together reflect olfactory notes that match together?

1. Goal and design

The project tests whether a color mapping of olfactory notes preserves the structure of real perfumery. It runs in two directions. **Part 1** takes existing perfumes (commercial vs niche), explains their notes and aroma-chemicals as a tree of accords, and derives a color palette from each perfume. **Part 2** checks whether notes that are used together have colors that sit close together, then generates new harmonious palettes and converts them back into perfume note structures.

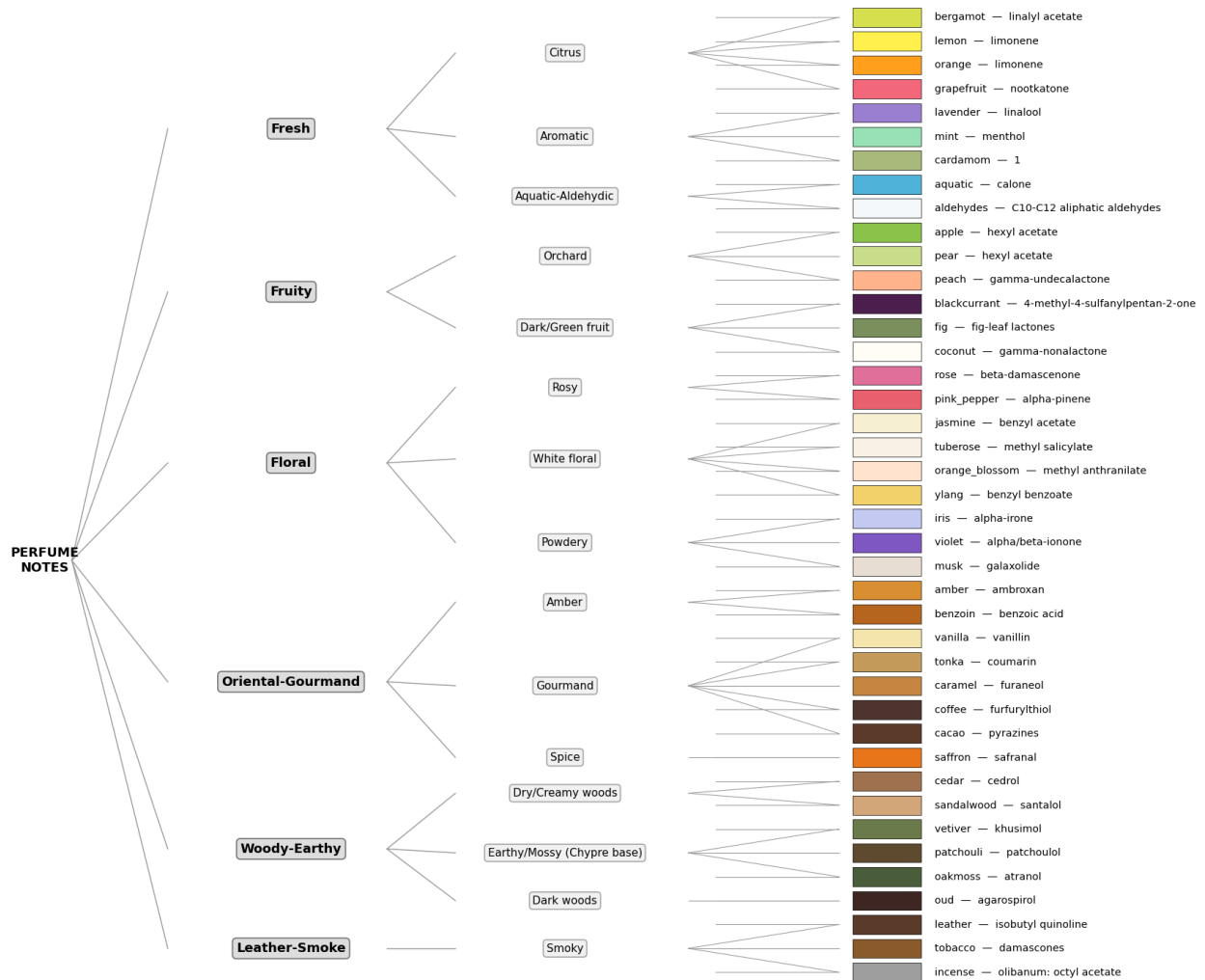
2. Important limitations (read first)

- **No live data.** The sandbox had no internet; Fragrantica has no public API and forbids scraping. The analysis uses a hand-curated **seed set of 40 perfumes** (20 commercial, 20 niche) with note lists written from memory. They are approximate. *ingest.py* sketches the legitimate route (Kaggle datasets, Reddit search JSON/PRAW) for scaling up.
- **Colors are hypotheses.** Hex codes are exact numbers but come from my reading of the odor-color literature (e.g. Levitan 2014; Schifferstein and Tanudjaja 2004), not from measurements. Validation requires a human cross-modal survey.
- **Circularity.** Colors were assigned partly by note family, so some association with note co-use is built in. The tests therefore compare against shuffled note-to-color mappings.
- **Small n.** With 40 perfumes the statistics only demonstrate the method.

3. Part 1: Olfactory tree and chemistry

Notes are organised as macro-family, accord, note, with a key aroma-chemical for each and the hypothesised color swatch. Classic accords of accords (Chypre = bergamot + mossy/earthy + rosy floral; Fougere = lavender + oakmoss + coumarin; Oriental = amber + gourmand + spice + woods) are encoded in data.py.

Olfactory tree: macro-family → accord → note (swatch = hypothesised synesthetic color; label = key molecule)



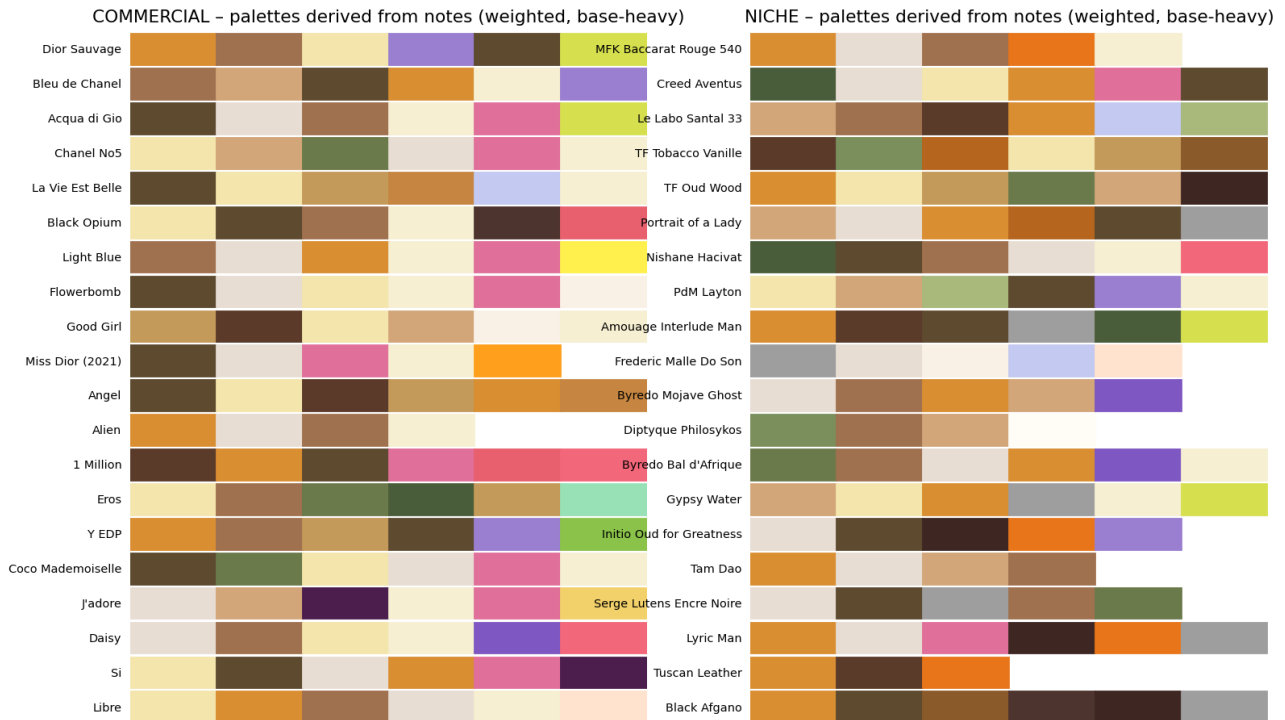
Note, color and chemistry table

Note	Hex	Family / accord	Key molecules	Facet
bergamot	#D6E04F	Fresh / Citrus	linalyl acetate, limonene, linalool	bright, green-citrus
lemon	#FFF04D	Fresh / Citrus	limonene, citral	sharp, sparkling
orange	#FF9F1C	Fresh / Citrus	limonene, sinensal	juicy, sweet citrus
grapefruit	#F2677A	Fresh / Citrus	nootkatone, 1-p-menthene-8-thiol	bitter, rosy-citrus
aldehydes	#F4F8FB	Fresh / Aquatic-Aldehydic	C10-C12 aliphatic aldehydes	soapy, champagne-like sparkle
aquatic	#4FB3D9	Fresh / Aquatic-Aldehydic	calone, floralozone	marine, ozonic
mint	#98E0B5	Fresh / Aromatic	menthol, carvone	cool, green
lavender	#9A7FD1	Fresh / Aromatic	linalool, linalyl acetate, coumarin	herbal-floral, clean
cardamom	#A9B97C	Fresh / Aromatic	1,8-cineole, alpha-terpinyl acetate	fresh spice
pink_pepper	#E8606E	Floral / Rosy	alpha-pinene, phellandrene, limonene	rosy, airy spice
saffron	#E8751A	Oriental-Gourmand / Spice	safranal	leathery, metallic-warm
apple	#8BC34A	Fruity / Orchard	hexyl acetate, ethyl 2-methylbutyrate	crisp, green-fruity
pear	#C8DC8A	Fruity / Orchard	hexyl acetate, ethyl decadienoate	juicy, watery fruit
peach	#FFB38A	Fruity / Orchard	gamma-undecalactone	velvety, creamy fruit
blackcurrant	#4B1E4D	Fruity / Dark/Green fruit	4-methyl-4-sulfanylpentan-2-one	tart, catty-fruity
fig	#7A8F5C	Fruity / Dark/Green fruit	fig-leaf lactones, stemone	green, milky-woody
coconut	#FFFDF5	Fruity / Dark/Green fruit	gamma-nonolactone	creamy, lactonic
rose	#E0709A	Floral / Rosy	beta-damascenone, citronellol, phenylethyl alcohol	petal, honeyed
jasmine	#F6EFD2	Floral / White floral	benzyl acetate, indole, hedione	indolic, radiant
tuberose	#FAF1E6	Floral / White floral	methyl salicylate, benzyl benzoate	narcotic, creamy
orange_blossom	#FFE3CF	Floral / White floral	methyl anthranilate, nerolidol	honeyed, soapy
ylang	#F2D16B	Floral / White floral	benzyl benzoate, germacrene D	banana-custard floral
iris	#C3C9F0	Floral / Powdery	alpha-irone	powdery, buttery, cool
violet	#7E57C2	Floral / Powdery	alpha/beta-ionone	powdery, sweet-green
musk	#E8DDD3	Floral / Powdery	galaxolide, ethylene brassylate, muscone	skin-like, clean
vanilla	#F3E5AB	Oriental-Gourmand / Gourmand	vanillin, ethylvanillin	sweet, creamy
tonka	#C49A5A	Oriental-Gourmand / Gourmand	coumarin	hay, almond
caramel	#C68642	Oriental-Gourmand / Gourmand	furaneol, maltol	burnt sugar
coffee	#4E342E	Oriental-Gourmand / Gourmand	furfurylthiol, pyrazines	roasted, bitter
cacao	#5B3A29	Oriental-Gourmand / Gourmand	pyrazines, theobromine	dark, bitter-sweet
amber	#D98E32	Oriental-Gourmand / Amber	ambroxan, labdanum, ambrox	warm, resinous, salty-skin
benzoin	#B5651D	Oriental-Gourmand / Amber	benzoic acid, cinnamates	balsamic, sweet
incense	#9E9E9E	Leather-Smoke / Smoky	olibanum: octyl acetate, alpha-pinene	smoky, church-like

Note	Hex	Family / accord	Key molecules	Facet
cedar	#A0714F	Woody-Earthy / Dry/Creamy woods	cedrol, Iso E Super	dry, pencil-wood
sandalwood	#D2A679	Woody-Earthy / Dry/Creamy woods	santalol, Javanol	creamy, milky wood
vetiver	#6B7A4B	Woody-Earthy / Earthy/Mossy (Chypre base)	khusimol, vetiverol	earthy, rooty, smoky
patchouli	#5D4A2F	Woody-Earthy / Earthy/Mossy (Chypre base)	patchoulol, norpatchoulol	damp earth, camphor
oakmoss	#4A5D3A	Woody-Earthy / Earthy/Mossy (Chypre base)	atranol, evernyl	mossy, forest floor
oud	#3E2723	Woody-Earthy / Dark woods	agarospirol, jinkoh-eremol	resinous, medicinal-animalic
leather	#5A3A29	Leather-Smoke / Smoky	isobutyl quinoline, birch tar	smoky-animalic
tobacco	#8B5A2B	Leather-Smoke / Smoky	damascones, megastigmatrienone	sweet hay-honey, dried leaf

4. Part 1: Palettes derived from perfumes

Each perfume's notes become a palette (top notes weight 1.0, heart 1.5, base 2.0; the six heaviest notes are shown). Lists were taken from the seed set.



Commercial vs niche

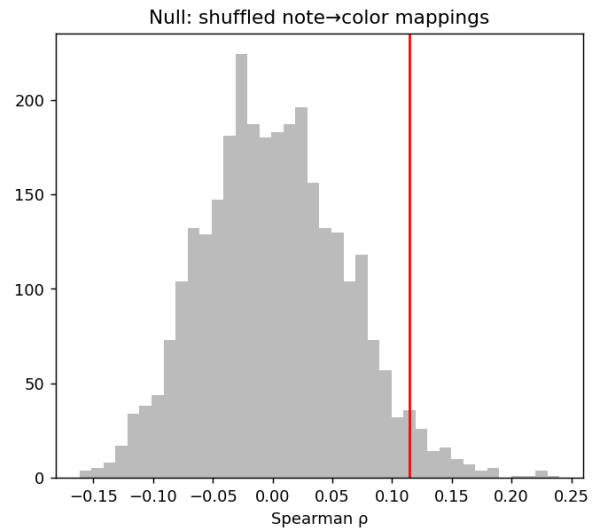
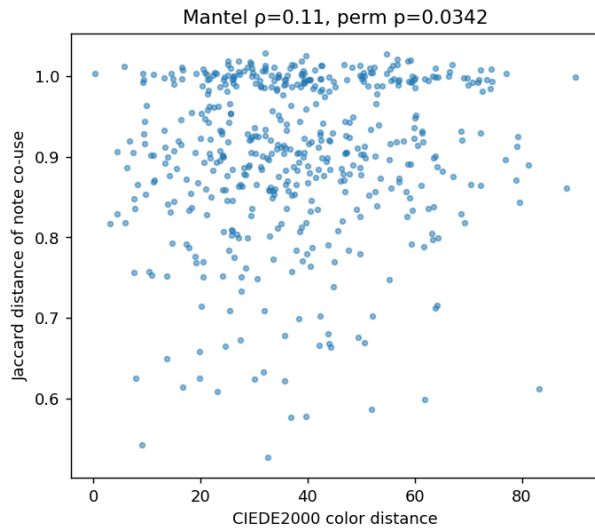
Metric	Commercial	Niche	Cohen's d	p (Mann-Whitney)
Notes per perfume	7.65	6.45	0.66	0.069
Mean lightness L*	69.64	63.23	0.77	0.015
Mean chroma C*	37.03	34.97	0.24	0.636
Palette contrast (mean dE2000)	34.05	31.30	0.49	0.148
Hue concentration (0-1)	0.76	0.80	-0.27	0.298
Families per perfume	3.85	3.70	0.18	0.736

- Commercial palettes are significantly **lighter** (L* 69.6 vs 63.2, p = 0.015) and tend to use more notes (7.7 vs 6.5, p = 0.07).
- Niche over-indexes on: tobacco, benzoin, saffron, oud, incense (incense 7 vs 0, oud and saffron 4 vs 0). Commercial over-indexes on: mint, pear, jasmine, ylang, caramel (jasmine 14 vs 5).
- Other differences (chroma, contrast, hue concentration, family count) are small and not significant at this sample size.

5. Part 2: Do matching colors reflect matching notes?

For notes appearing in at least 3 perfumes, I compared color distance (CIEDE2000) with note distance (Jaccard distance of the perfumes in which they appear) using a Mantel-style Spearman test. The null distribution comes from 10,000 random relabelings of which color belongs to which note.

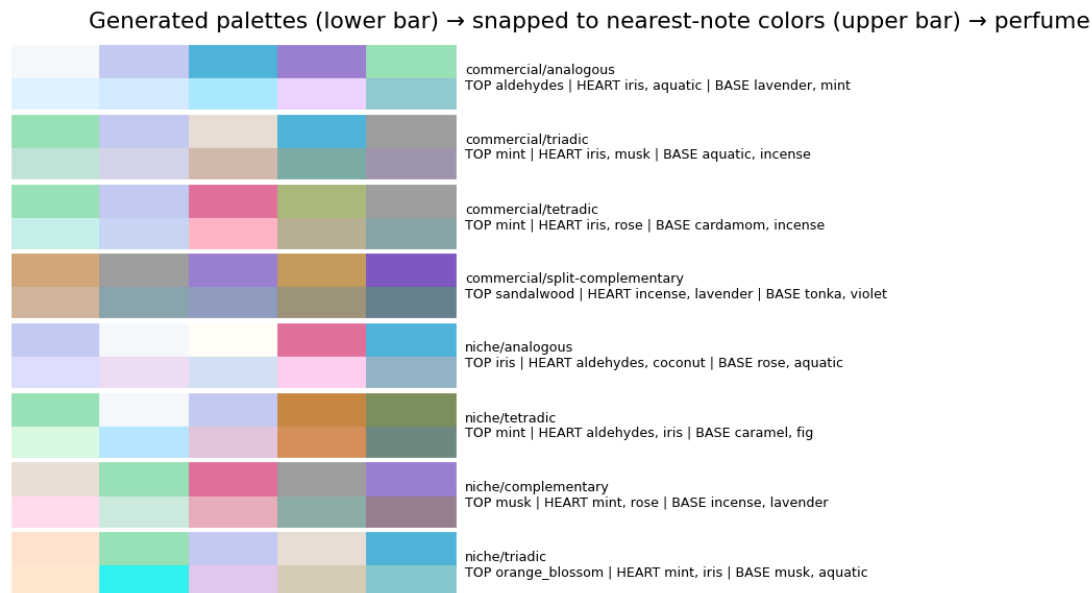
Subset	Notes	Pairs	Spearman rho	Permutation p
all	32	496	0.114	0.034
commercial	27	351	0.069	0.153
niche	27	351	0.070	0.148



- Pooled: $\rho = 0.11$, $p = 0.034$, a small marginal effect. Within each segment alone it is not significant (p about 0.15).
- Notes in the same family are closer in color: mean dE2000 32.7 within vs 37.3 between families ($p = 0.002$). Partly built in by construction.
- Hypothesis that base notes have darker colors: not supported ($\rho = -0.25$, $p = 0.13$, wrong sign).

6. Part 2: Generating palettes and perfumes

Method. (1) Generate 6,000 candidate 5-color palettes from harmony schemes in LCh space (analogous, complementary, triadic, split-complementary, tetradic). (2) Score each with a logistic-regression critic trained on real perfume palettes vs random palettes, using 7 features (mean and sd of pairwise dE2000, mean and sd of L*, mean and sd of chroma, hue concentration). (3) Keep the top diverse palettes per segment. (4) Snap each color to the nearest note by CIEDE2000. (5) Assign lightest color to top notes, darkest to base notes.



Segment	Scheme	Top	Heart	Base	PMI compat.
commercial	analogous	aldehydes	iris, aquatic	lavender, mint	-0.68
commercial	triadic	mint	iris, musk	aquatic, incense	-0.86
commercial	tetradic	mint	iris, rose	cardamom, incense	-0.68
commercial	split-complementary	sandalwood	incense, lavender	tonka, violet	-0.80
niche	analogous	iris	aldehydes, coconut	rose, aquatic	-0.10
niche	tetradic	mint	aldehydes, iris	caramel, fig	-0.16
niche	complementary	musk	mint, rose	incense, lavender	-0.86
niche	triadic	orange_blossom	mint, iris	musk, aquatic	-0.58

Honest assessment: the generator failed.

- Generated perfumes have mean note-compatibility (pointwise mutual information) of -0.59, **lower** than random note sets (-0.38) and far below real perfumes (0.17).
- The critic's cross-validated AUC (about 0.99) is inflated: random palettes are trivially different and augmented samples leak across folds.
- Generated palettes drift to low-chroma pastels, which snap poorly to notes (mean dE about 12), producing watery, fresh-heavy structures.

7. Conclusions and next steps

- Conclusion on the research question:** on this seed data there is only weak, marginal evidence (rho about 0.11) that color closeness tracks note compatibility, and part of that is built into the mapping. The data cannot yet confirm or reject a real color-smell link.

- **1. Real data:** 1,000+ perfumes via Kaggle Fragrantica-derived datasets (check licences) and Reddit mentions as a community-perception layer.
- **2. Human survey:** participants pick colors for notes from a CIELAB grid; use median responses as the color mapping; this removes the circularity.
- **3. Better generator:** grouped cross-validation, harder negatives, and joint optimisation of color harmony and PMI rather than snapping afterwards; also fit separate commercial and niche models on the larger data.

Code: data.py, run.py, ingest.py, make_pdf.py. Outputs: out/*.csv, out/results.json, out/fig1-4.png.